

SHRESTH SAMYAK

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EDUCATION

Thapar Institute of Engineering & Technology
B.E. — Computer Science & Engineering (DSAI)

2025 – 2029
Patiala, Punjab

EXPERIENCE

AI Automation Engineer

Apr 2026 – Present

GetHelpDesk.ai · Internship

Building an intelligent automation layer that takes structured patient and insurance data and executes the full data-entry workflow inside dental PMS software (Open Dental, Dentrrix) with high accuracy — eliminating manual data entry.

Engineered a coverage mapping engine that translates abstract insurance tiers (e.g. Basic = 80%) into ADA code-range configurations, feeding a deterministic pywinauto execution layer.

Implemented a state-machine JobOrchestrator with idempotency tracking, MySQL duplicate checking, and failure evidence collection (screenshots + audit logs).

Freelance Generative AI Engineer

Aug 2024 – Apr 2025

Fiverr

Delivered 15+ production-ready LLM chatbots (GPT-4, Claude), maintaining 95%+ client satisfaction; architected RAG pipelines reducing query latency by 60%.

Integrated chatbots with FastAPI and Telegram Bot API, supporting 1,000+ DAUs; led prompt A/B testing improving accuracy by 35% and cutting token spend by 25%.

Implemented deduplication, content hashing, and retry mechanisms achieving 99.5% uptime across ingestion workflows.

SELECTED PROJECTS

LLM Diet — Token Compression for AI Coding Agents

🐙 GitHub 🐍 PyPI

Built and published a PyPI package that intercepts Claude Code file reads via a shadow MCP server, returning AST call-graph-compressed versions instead of raw files — achieving 69% token reduction (32K → 10K chars) on real codebases.

Replaced RAG with deterministic AST-based call graph analysis: zero LLM calls at retrieval time, same query always returns the same result; benchmarked >99.9% token reduction on a 946-node FastAPI project.

Supports Claude Code (full shadow MCP), Cursor, and Windsurf; ships `context-engine` CLI for one-command codebase indexing and `diet-run` for enforced low-bandwidth mode.

Self-Correcting LLM — Critic–Refiner Loop with Tool-Augmented Grounding

🐙 GitHub

Fine-tuned Mistral-7B via QLoRA on a synthetically corrupted dataset; designed Generator–Critic–Refiner architecture benchmarked across four system configurations.

Trained a MiniLM critic for quality score regression and verdict classification; integrated browser-augmented grounding (DuckDuckGo + w3m) triggered on hallucination detection.

Documented failure modes: non-monotonic refinement, hallucination persistence, critic–reality misalignment, and stagnation.

Surveillance — Real-Time CCTV & Abandoned Bag Detection

🐙 GitHub

Fine-tuned YOLO on a custom railway CCTV dataset; implemented BoT-SORT with BagRegistry embedding similarity for appearance-based re-identification.

Engineered FSM-based ownership tracking and multi-factor abandonment risk scoring via AnomalyFusion Engine; achieved ~50ms/frame at 1280px on RTX 4060.

ChronoMerge — Testimony Intelligence System

AI pipeline (Whisper/Groq + LLaMA + Git-style conflict detection) transforming multi-witness testimonies into structured timelines with risk scoring and hallucination grounding.

Dual interfaces: Witness Portal and Investigator Dashboard (timeline viz, conflict graphs, AI copilot); FastAPI async backend with PostgreSQL and modular LLM orchestration.

TECHNICAL SKILLS

LLM & Fine-Tuning: QLoRA, LoRA, PEFT, Mistral-7B, MiniLM, Hugging Face, instruction tuning, evaluation pipelines

AI & ML: GPT-4, Claude, LLaMA, RAG, embeddings, prompt engineering, LangChain, LlamaIndex, YOLO, BoT-SORT, computer vision

Dev Tools & Infra: MCP servers, AST analysis, Claude Code, shadow MCP, context injection, Docker, Git, n8n

Automation: Windows UIA, Pywinauto, Win32 messaging, state-machine orchestration, idempotency systems

Infra & Backend: FastAPI, Flask, PostgreSQL, MySQL, Oracle Cloud, AWS, OpenAI API, Groq API, Whisper, OpenCV, Pydantic v2

Languages: Python, SQL, JavaScript **Vector DBs:** Pinecone, ChromaDB, FAISS

CERTIFICATIONS & AWARDS

Oracle Cloud Infrastructure — Certified Generative AI Professional (2025) Oracle Cloud Infrastructure — AI Foundations Associate (2025)

Winner — Feature Creep Hackathon (Nov 2025) Shortlisted — Outlier AI UI Design Hackathon (Apr 2025)